Virtual Event Scheduler and Management Platform

Software Requirements Specification

Introduction

This document defines the software requirements for the Virtual Event Scheduler and Management Platform. The system consists of a desktop application for staff and administrators, and a web application for attendees. Both are connected to a shared database via a RESTful Web API.

Functional Requirements

ID	Requirement	Description
FR-1	User Registration and Login	The system shall allow users to create an account by providing their full name, email address, and password. Login is performed using email and password credentials. Users shall be able to log in and log out.
FR-2	Role-Based Access Control	The system shall assign one of three roles to each user: Admin, Staff, or Attendee. Admins can manage users, roles, and all events. Staff can create, edit, and delete events and view participant lists. Attendees can view events, register for events, and receive notifications.
FR-3	Event Creation	Authorized users (Admin/Staff) shall be able to create new events with title, description, date, time, location, and capacity.
FR-4	Event Editing and Deletion	Authorized users shall be able to update or delete existing events.
FR-5	Event Viewing	Users shall be able to view the list of upcoming events.
FR-6	Event Registration	Users shall be able to register for available events.
FR-7	Participant Tracking	Admin/Staff users shall be able to view the list of participants registered for each event.
FR-8	Event Filtering Using LINQ	The system shall allow users to filter events based on date, availability, or category using LINQ queries.
FR-9	Notification System Using Delegates	The system shall send notifications to users when they register for events or when event details change.
FR-10	Event Status Management	Admin users shall be able to change event status (Active, Completed, Cancelled, Postponed).

Non-Functional Requirements

ID	Requirement	Description
NFR-1	Performance	The system shall load event lists within 2 seconds under normal conditions.
NFR-2	Security	All passwords shall be stored using bcrypt hashing (minimum cost factor 10). All API endpoints shall require JWT-based authentication. Communication shall be encrypted via HTTPS/TLS 1.2 or higher. Access tokens shall expire after 60 minutes of inactivity.
NFR-3	Usability	The system shall achieve a System Usability Scale (SUS) score of at least 75. Every page shall include a Help button that opens context-sensitive guidance for that screen. All error messages shall be written in plain language without technical jargon.
NFR-4	Reliability	The system shall auto-save user data every 60 seconds. In the event of a power outage or unexpected shutdown, no more than 60 seconds of data entry shall be lost.
NFR-5	Scalability	The system shall support a minimum of 5,000 registered users and 500 concurrent sessions without performance degradation. The database shall handle at least 10,000 event records.

Scenario

Aslan opens the web application and creates an account with his name, email, and password. The system gives him the Attendee role.

He browses upcoming events and filters them by date and category. He finds a workshop and registers for it. The system immediately sends him a confirmation notification.

A staff member logs in through the desktop application and creates a new workshop with a title, description, date, time, and capacity. Later, he notices a mistake in the date and edits the event. All registered participants, including Aslan, receive a notification about the change.

An admin opens the participant list for the workshop to check registrations. The event fills up, so he changes its status to “Active – Full.” After the workshop ends, he updates the status to “Completed.”

Throughout the process, all passwords are hashed and connections are encrypted. Every page loads quickly and includes a Help button. Data is auto-saved every 60 seconds in case of any interruption. The platform handles thousands of users at the same time without slowing down.